# Does Topic Sentiment Cause Perceived Ideology?

Comparing Human and LLM Annotations in Political News Articles

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## Motivation

LLMs are widely used as low-cost stand-ins for human annotators. But matching human *outputs* need not mean matching the *reasoning* behind them.

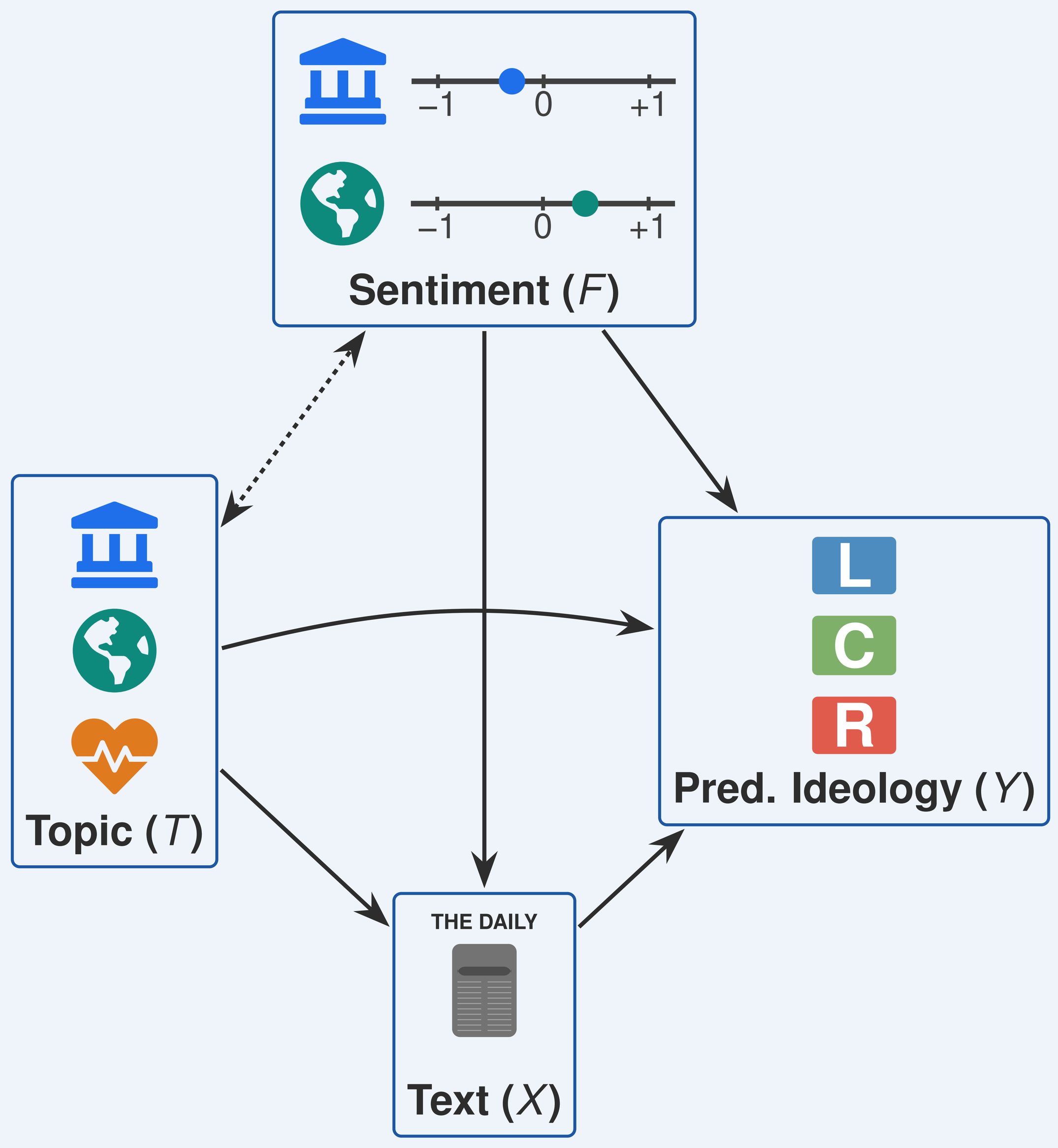
We test **causal fidelity**: do the *same features* drive LLM and human labels? Accuracy and agreement compare *what* is predicted, never *why*.

**Research question**

Does *topic sentiment* causally shift *perceived ideology*, and does the answer depend on *who* assigns the label?

## Causal Framework

Hold sentiment **fixed**, vary only the ideology label: any difference in the estimated effect is attributable to the *annotator*.

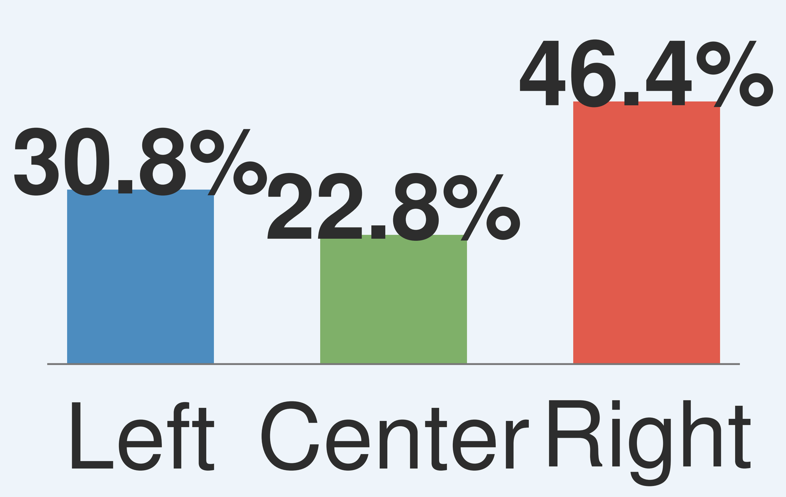



- *X* **Text**: the full article text.
- *T* **Topic tags** *(confounder)*: which subjects the article covers (e.g. immigration, climate).
- *F* **Sentiment** *(treatment)*: per-topic sentiment, scored from −1 to +1.
- *Y* **Ideology** *(outcome)*: the predicted Left / Center / Right label.

## Experimental Setup

AllSides™

Expert editors rate each article **Left**, **Center**, or **Right**. Corpus: **1,265** articles (Baly et al., 2020).



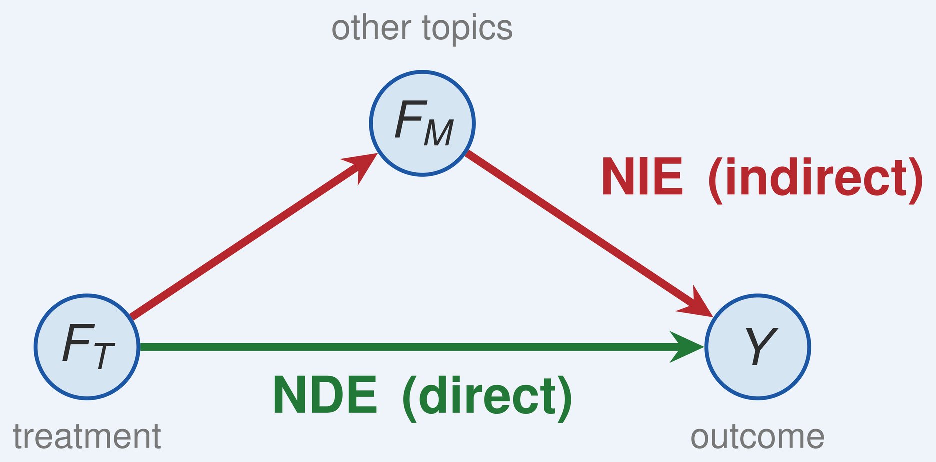

Human (AllSides) ideology split

**Sentiment (*F*) is held fixed**: Llama scores per-topic sentiment from −1 to +1. Only the **ideology label** varies across four annotators:

- **Human** (AllSides experts)
- **GPT-4o-mini**, base
- **GPT-4o-mini**, fine-tuned
- **Llama-3.3-70B**

## Estimands

Topic tags cluster into co-occurrence **communities** (Louvain). Per community, `LinearDML` with bootstrap CIs ($B$=2,000) estimates the **ATE** (overall effect of community sentiment on ideology) and a **mediation** split of the total effect, TE = NDE + NIE:




A near-total **NDE** means sentiment maps to ideology through a **direct** edge, not via other mediator topics.

**Sign convention.** A **negative** estimate means positive sentiment predicts a *Left*-leaning label; a **positive** estimate means positive sentiment predicts a *Right*-leaning label.

## Topic Communities

Three sample communities, used throughout the results below:

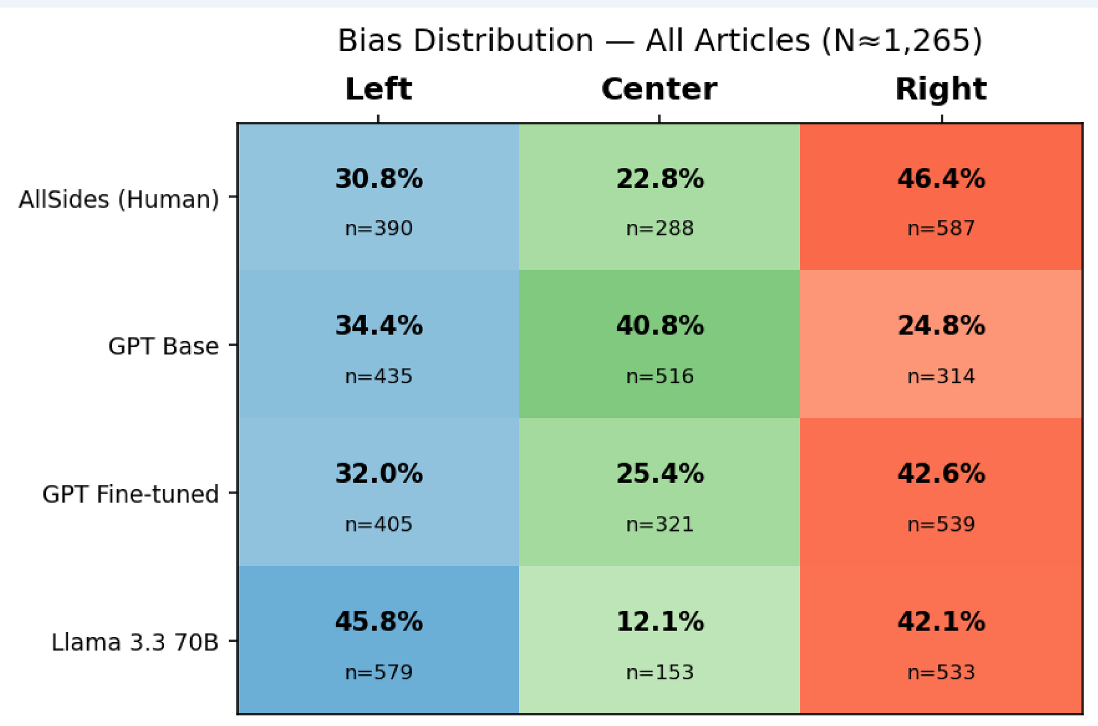
- **C11 Trump Presidency**: Jan. 6, Trump, impeachment
- **C3 Social Issues**: immigration, LGBTQ, abortion
- **C9 Geopolitics & Climate**: foreign policy, climate

## Accuracy Hides the Coupling

We score each LLM's ideology labels against the human experts (macro-F1). Fine-tuning lifts GPT from 50 to 72, but F1 says nothing about *which features* a model relies on.

| Annotator | F1 |
|---|---|
| GPT-4o-mini *(fine-tuned)* | **72.48** |
| Llama-3.3-70B | 54.61 |
| GPT-4o-mini *(base)* | 50.07 |

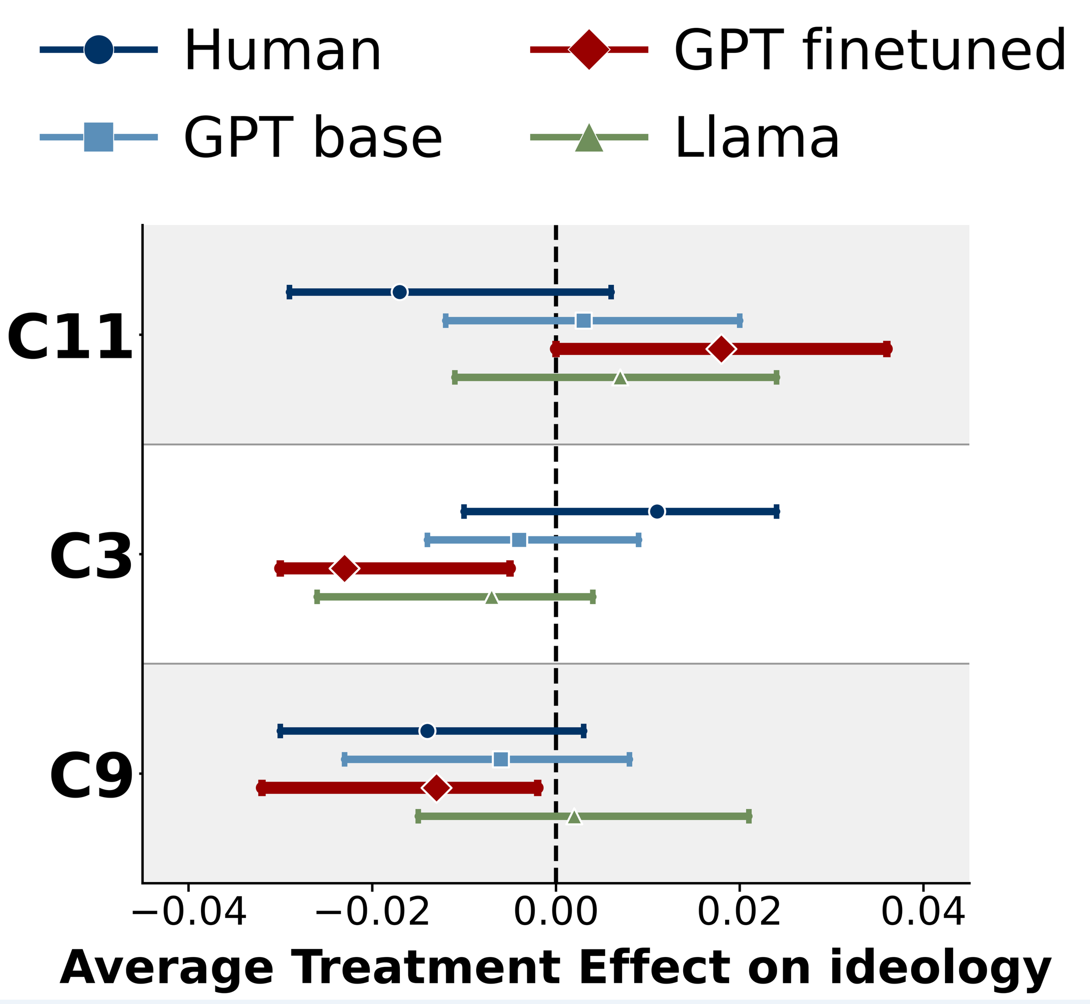
All four annotators label the *same* articles, yet their Left/Center/Right distributions diverge sharply:




Base GPT over-predicts **Center**; Llama leans **Left**; the fine-tuned GPT model tracks the human **Right**-leaning split most closely. Same inputs, different label-generating behaviour, which only a causal analysis can separate.

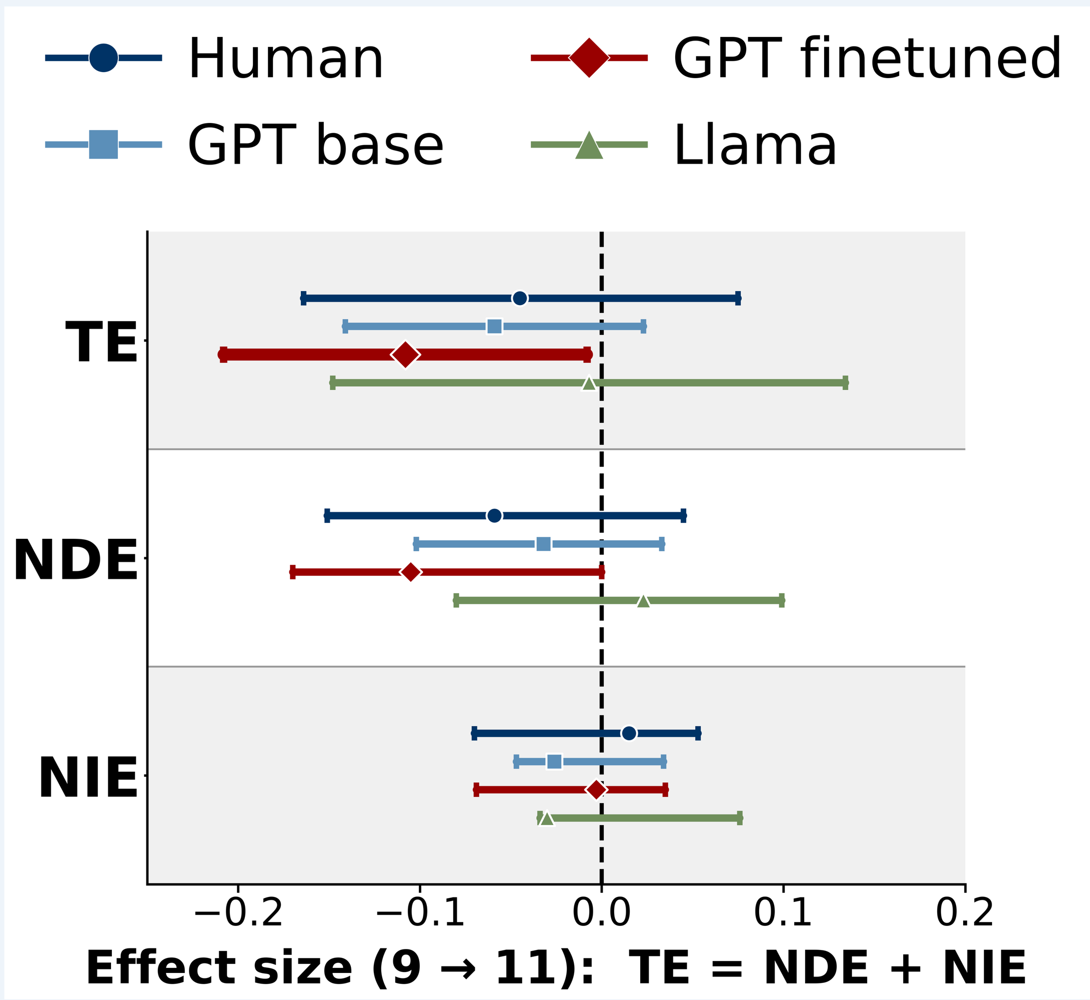
**Highest F1 ≠ faithful causal structure.**
The most accurate model still hides spurious sentiment couplings.

## Result 1: Community-level Effects




- **Human and Llama**: no significant effects.
- **Fine-tuned GPT**: the *only* model with significant community effects.
- Fine-tuning can **flip the effect's sign** relative to humans.

## Result 2: Mediation (C9 → C11)




Geopolitics/climate sentiment → ideology. **Only fine-tuned GPT is significant:**

TE = −0.108 | NDE = −0.105 | NIE ≈ 0

The effect is **almost entirely direct** (≈3% mediated): framing of geopolitics related topics map straight to the ideology label and aren't mediated through sentiment of Trump presidency related topics.

## Takeaways

**Fine-tuning, model family, and topic *interact*.**
Together they set how prominent the sentiment→ideology pathway is, and whether that pathway *aligns* with expert human labellers, not uniformly better or worse.

**Reversed** **C11 Trump Presidency**: human annotators show a *Left*-leaning effect, yet *every* LLM flips the sign to *Right*.

**Farther** **C3 Social Issues**: fine-tuning adds a spurious coupling, *reversing* the sentiment effect's sign relative to humans.

**Closer** **C9 Geopolitics & Climate**: a muted zero-shot effect is strengthened into *agreement* with the human estimate.

- Higher F1 does **not** guarantee faithful causal structure.
- Judge silver labels by **causal alignment** per domain, not accuracy.

**Key references.** Pearl (2001), *Direct and indirect effects*; Chernozhukov et al. (2018), *Double Machine Learning*; Gao et al. (2025), *LLMs as cautionary surrogates*; Baly et al. (2020), *AllSides news corpus*.

Code: `causal-inference-on-text`

Data: `AllSides-sentiments`